Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_36

1. Summary of Key Deviations

The analysis of Run_36 reveals a catastrophic failure of the entire fuel system, characterized by a severe and system-wide loss of flow and pressure. However, there is a contradictory positive deviation in combustion pressure.

- **Flow_Pump (Major):** The primary and most critical anomaly is a massive **negative offset and drift**, with pump flow dropping by a staggering **-55.1%** compared to a healthy run. This indicates a near-total failure of the pump to deliver fuel.
- **Flow_Bypass (Critical):** This signal shows a **critical dropout**, falling by **-97.9%**. This confirms that there is almost no flow being produced by the pump to be bypassed.
- **System-Wide Pressure & Flow Loss (Moderate to Major):** Almost every other pressure and flow sensor in the system (`p_Pump`, `p_FMU`, `p_Shut`, `Flow_Engine1-4`, `Flow_PRV`) shows a major, corresponding **negative offset**, with readings dropping by **-10% to -40%**.
- **p_Combustion (Anomalous):** In stark contrast to every other reading, `p_Combustion` shows a minor **positive offset** of approximately **+6%**. This is physically inconsistent with a system that is being starved of fuel.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Main Fuel Pump & p_Combustion Sensor**	**Dual Fault: Catastrophic Pump Failure & Sensor Calibration Error**	This is the only hypothesis that fits all the evidence. There are two independent failures: 1) The **Main Fuel Pump has catastrophically failed**, causing a system-wide loss of flow (-55.1%) and pressure. 2) The **p_Combustion sensor has a positive offset calibration error**, causing it to read high even when the real pressure is low.

2	**Main Fuel Pump**	**Catastrophic Failure / Seizure**	This explains the primary physical fault: the massive drop in `Flow_Pump`, `Flow_Bypass`, and almost all system pressures. However, it does **not** explain why `p_Combustion` reads high when it should be reading very low.
3	**Boost-Pressurised Tank / Inlet Line**	**Complete Blockage / Clogged Filter**	This would cause symptoms nearly identical to a failed pump (a major drop in flow and pressure). This is a strong candidate for the physical part of the fault, but like hypothesis #2, it does not explain the anomalous `p_Combustion` reading.
4	**Catastrophic Leak (Post-Pump) & p_Combustion Sensor Fault**	**Dual Fault: Major Leak & Sensor Error**	A major leak could cause the flow and pressure drop, and a separate sensor fault could explain the `p_Combustion` reading. However, the `Flow_Pump` reading is so low that it's more indicative of a pump failure than a leak.
5	**Multiple Simultaneous Sensor Failures**	**System-wide Sensor Errors**	It is astronomically unlikely that every single flow and pressure sensor would fail simultaneously in a physically consistent way (all negative, except for one positive), pointing to a real physical event being the primary driver.

3. Most Probable Hypothesis

The most probable root cause is a **Dual Fault Scenario: (1) a Catastrophic Failure of the Main Fuel Pump (or a total inlet blockage) AND (2) a concurrent, independent Positive Offset Failure of the p_Combustion Sensor.**

Reasoning: The data clearly shows two distinct and contradictory problems. The first is a catastrophic loss of flow and pressure throughout the system, which unequivocally points to a failure of the Main Fuel Pump or a total blockage of its

inlet. The second is a `p_Combustion` reading that is physically impossible given the first fault—pressure cannot be high when there is no fuel flow. This proves that in addition to the catastrophic physical failure, the `p_Combustion` sensor has also failed with a positive offset error, causing it to report a high pressure value regardless of the real-world condition.